

Module 7: Assignment

Security-usability design choices

MCY 630 Security Architecture

Assignment Submission Instructions

Submission requirements: assignment

File format: MSWord or pdf (preferred)

Report requirements

- Include your name and assignment title at the top of the document.
- Number the questions and answers and answer each question in your submission in sequence.
- For each response, include the required text as well as any necessary images and diagrams.
- Use black font color for texts
- If uploading an MSWord format file, view the uploaded file from Canvas to make sure that the layout of images and texts are not overlapping in the report. If the contents seem to have moved around, upload a pdf instead of an MSWord file.

Scenario

Consider the application scenario from Module 2 Assignment:

A simple online food pick-up ordering application platform for a restaurant, MCY Fast Food, allows online users to perform the following actions using the web-based application interface:

- Browse through the restaurant's menu options, corresponding prices, and nutrition information.
- Add items using a "add" button to the cart.
- View the cart to finalize the order.
- Items can be deleted from the cart before proceeding to finalize the order.
- Provides all necessary personal user information and pays the total amount due using a credit card gateway.
- Receive confirmation email with order details, itemized bill, and approximate pick-up time.
- The restaurant also receives an email with the order details, itemized bill, and approximate pick-up time for the customer.

Questions

1. (15 points) Discuss the usability and security aspects for the above application scenario based on the Security-Usability Threat Model.
2. (15 points) Based on the assessment techniques applied for Security-Usability Threat Model, discuss the probable de-motivators where users may unknowingly end up breaking the system.
3. (15 points) Based on the assessment techniques applied for Security-Usability Threat Model, discuss the probable motivators where users may end up performing threat scenarios due to the secure methods being more complicated compared to the regular usage scenarios.
4. (5 points) Discuss, in your opinion, the choices for triggers, warnings, and indicators which can be adopted for the application interface.